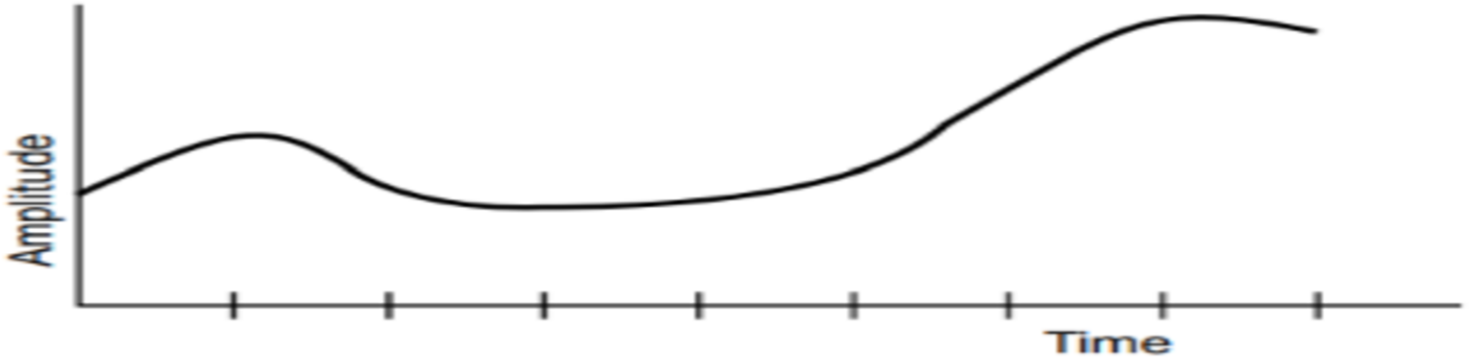
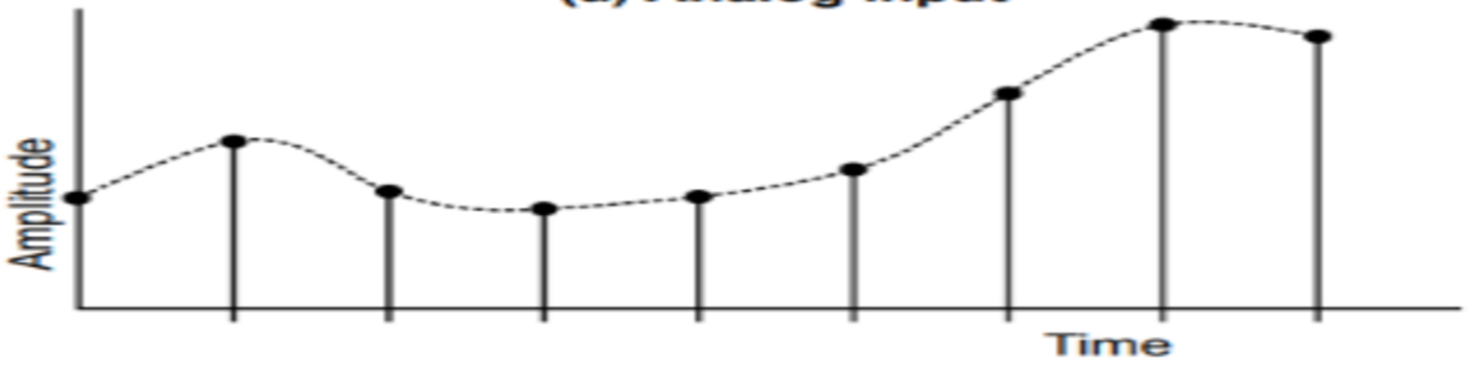
Sample and Hold Circuit

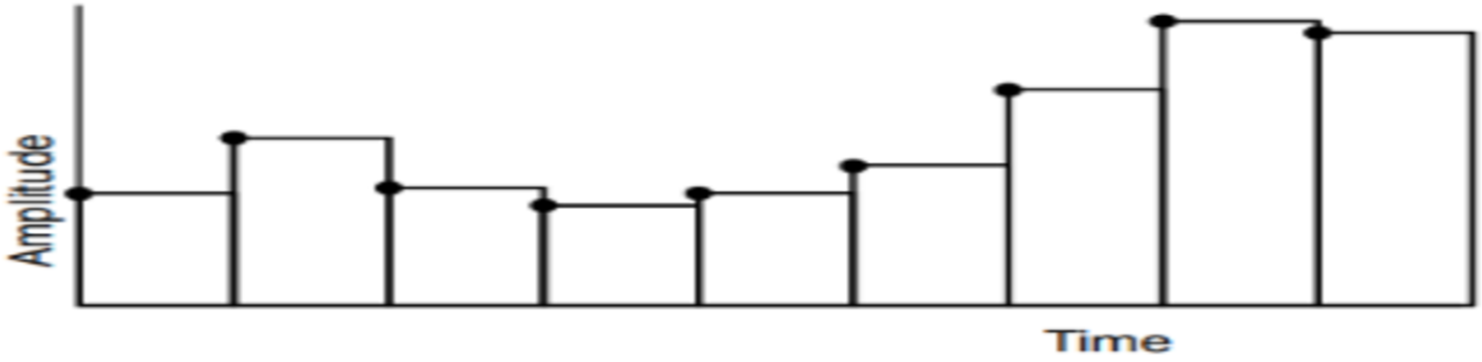
- As its name implies, the sample and hold (S/H) circuit samples the value of the input signal in response to a sampling command and hold it at the output until arrival of the next command.
- It samples an analog input voltage in a very short period, generally in the range of 1 to 10 ps, and holds the sampled voltage level for an extended period, which can range from a few millisecond to several seconds.



(a) Analog input

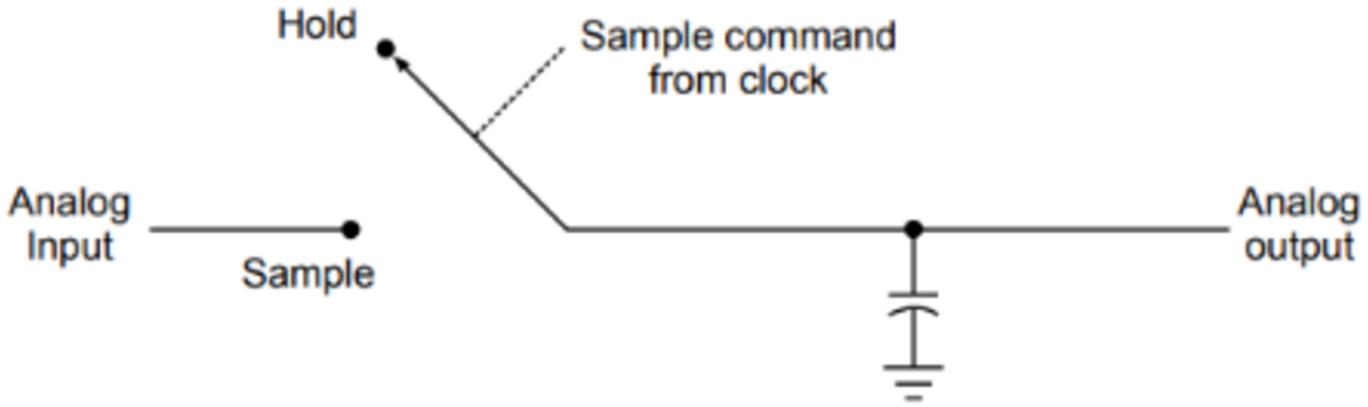


(b) Sampled input

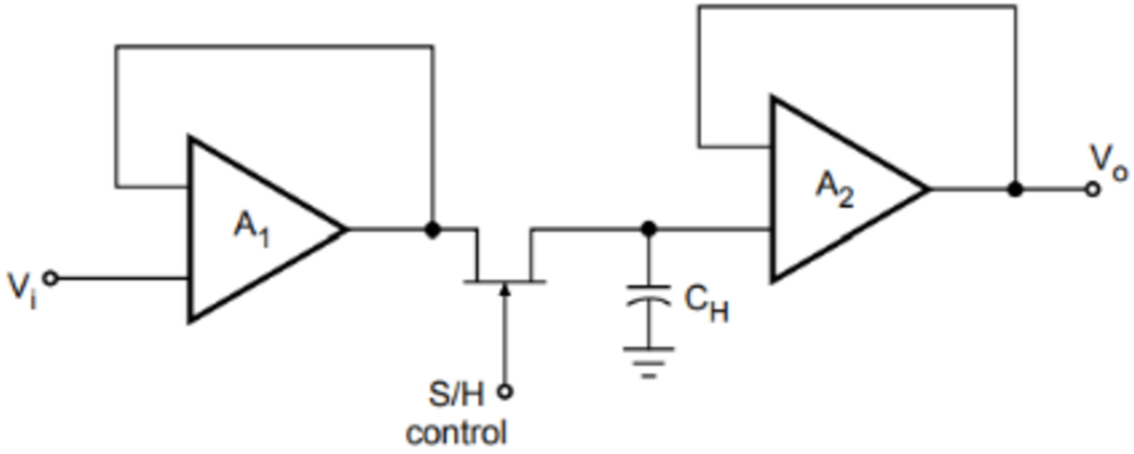


(c) Analog hold input

- The sample and hold circuit uses to basic components analog switch and capacitor. Figure 3 shows the basic sample and hold circuit.

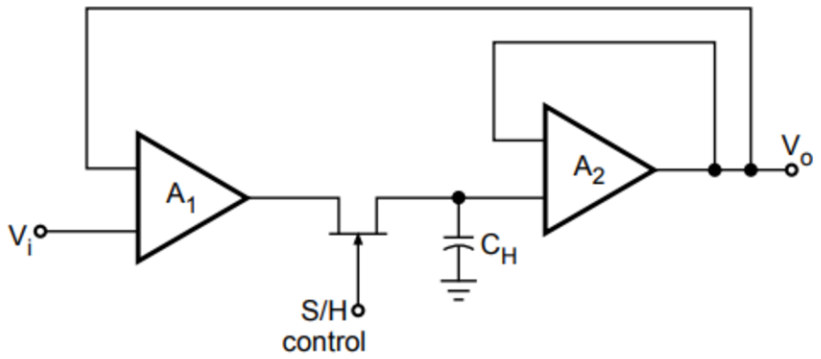


- The circuit tracks the analog signal until the sample command causes the digital switch isolate the capacitor from to the signal, and the capacitor holds this analog voltage during A/D conversion.

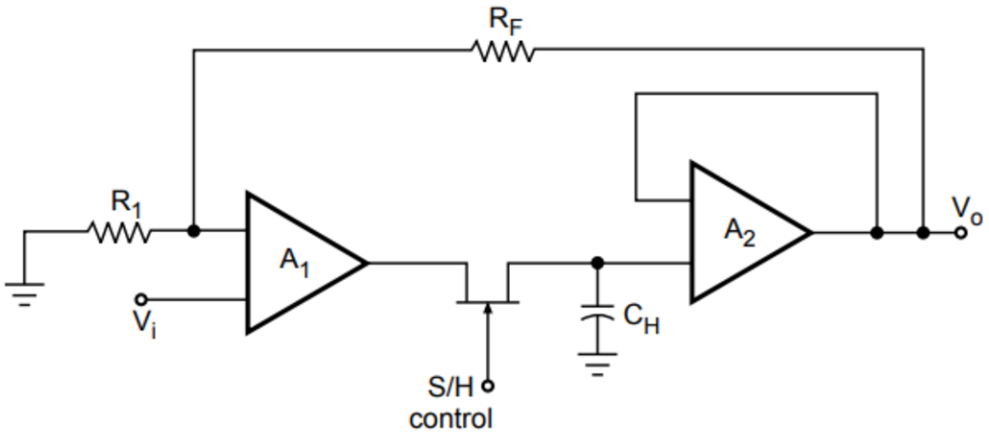


- Here JFET is used as switch.
- During the sampling time the JFET switch is turned on, and the holding capacitor charges up to the level of the analog input voltage.
- At the end of this short sampling period, the JFET switch is turned off. This isolates the holding capacitor C_H from the input signal. As a result, the voltage across capacitor C_H and hence the output voltage will remain essentially constant at the value of the input voltage at the end of the sampling time.
- However, there will be a small drop-off or drop of the capacitor voltage during the hold period due to the various leakage currents. To avoid this, input and output buffers (voltage follower) circuits are used.

- The acquisition time of a S/H circuit is the time required for the holding capacitor C_H to charge up to a level close to the input voltage during sampling. The acquisition time for S/H circuit should be as low as possible. In the circuit of Fig.(a) there are three principle factors that will control the acquisition time. These factors are :
- RC time constant where R is the r_{ds} (ON), i.e. on resistance of JFET and C is the holding capacitance C_H
- Maximum output current, which can be source or sunk by the operational amplifier.
- Slew rate of the op-amp.



The circuit shown in Fig. (b) offers some advantage over that of Fig. (a) in terms of the acquisition time since the rds (ON) of the JFET switch is inside the feedback loop of A_1 and A_2. Therefore the acquisition time for this circuit is limited by maximum output current and slew rate of the op-amp, rather than the RC time constant.



(c)

The S/H circuit of Fig. (c) performs in a fashion similar to that of Fig.(b) but it offers the additional feature of providing voltage gain.

Voltage gain of circuit given as $A = 1 + (R_F/R_1)$.

Therefore, the sampled output voltage is equal to the sampled input voltage multiplied by the voltage gain factor $1 + (R_F/R_1)$.